



POSIFY

Internship Project at **Asteroidea**

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Academic Year: 2025 - 2026



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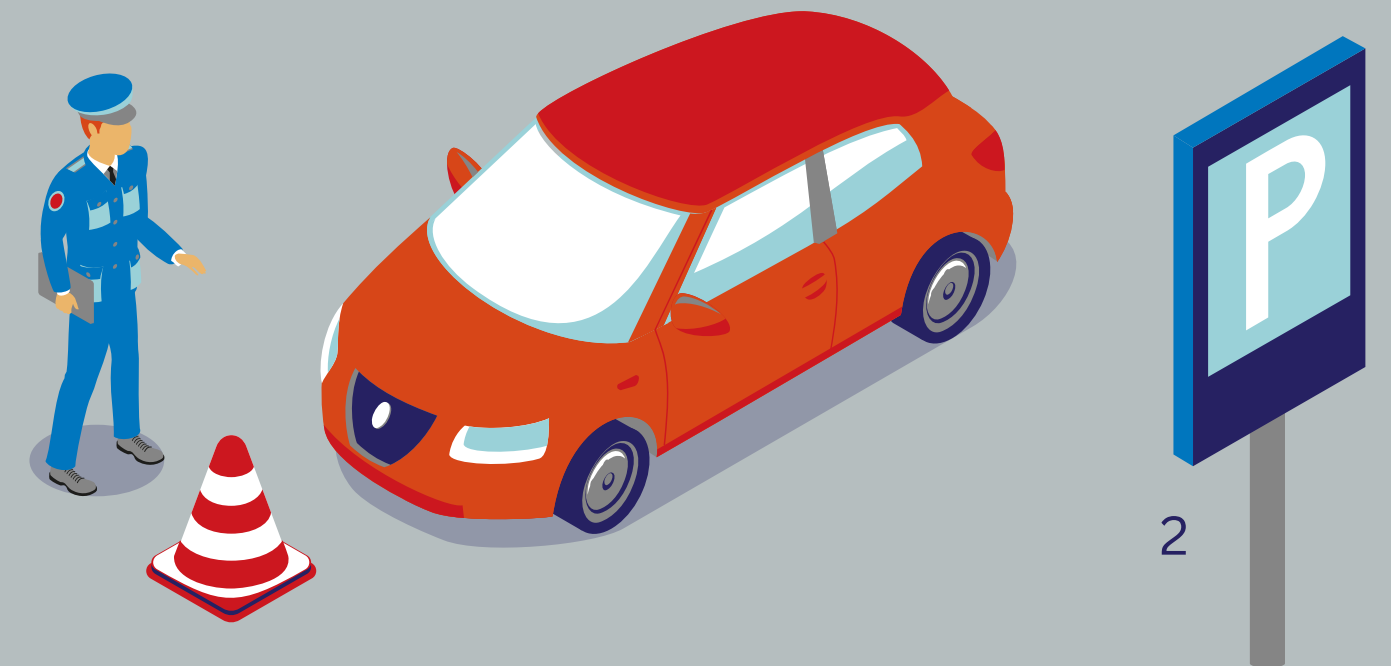
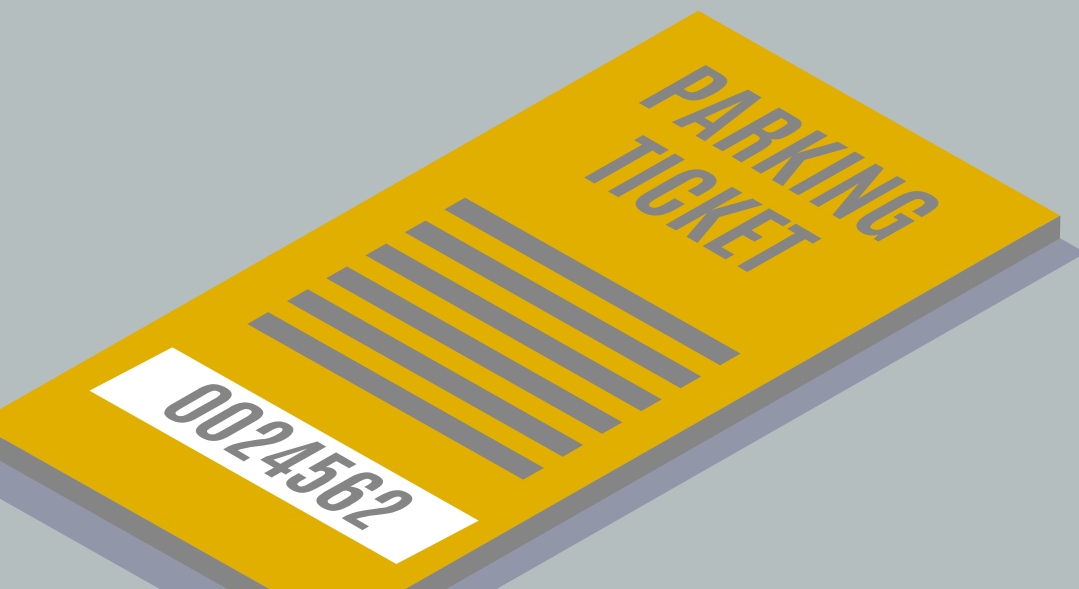
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01

Introduction

Why Asteroidea?

Rare Stack

Golang in production: almost unheard of in Tunisia



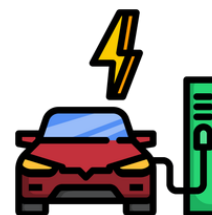
Career Fit

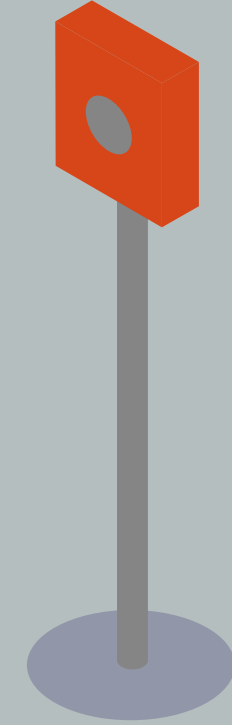
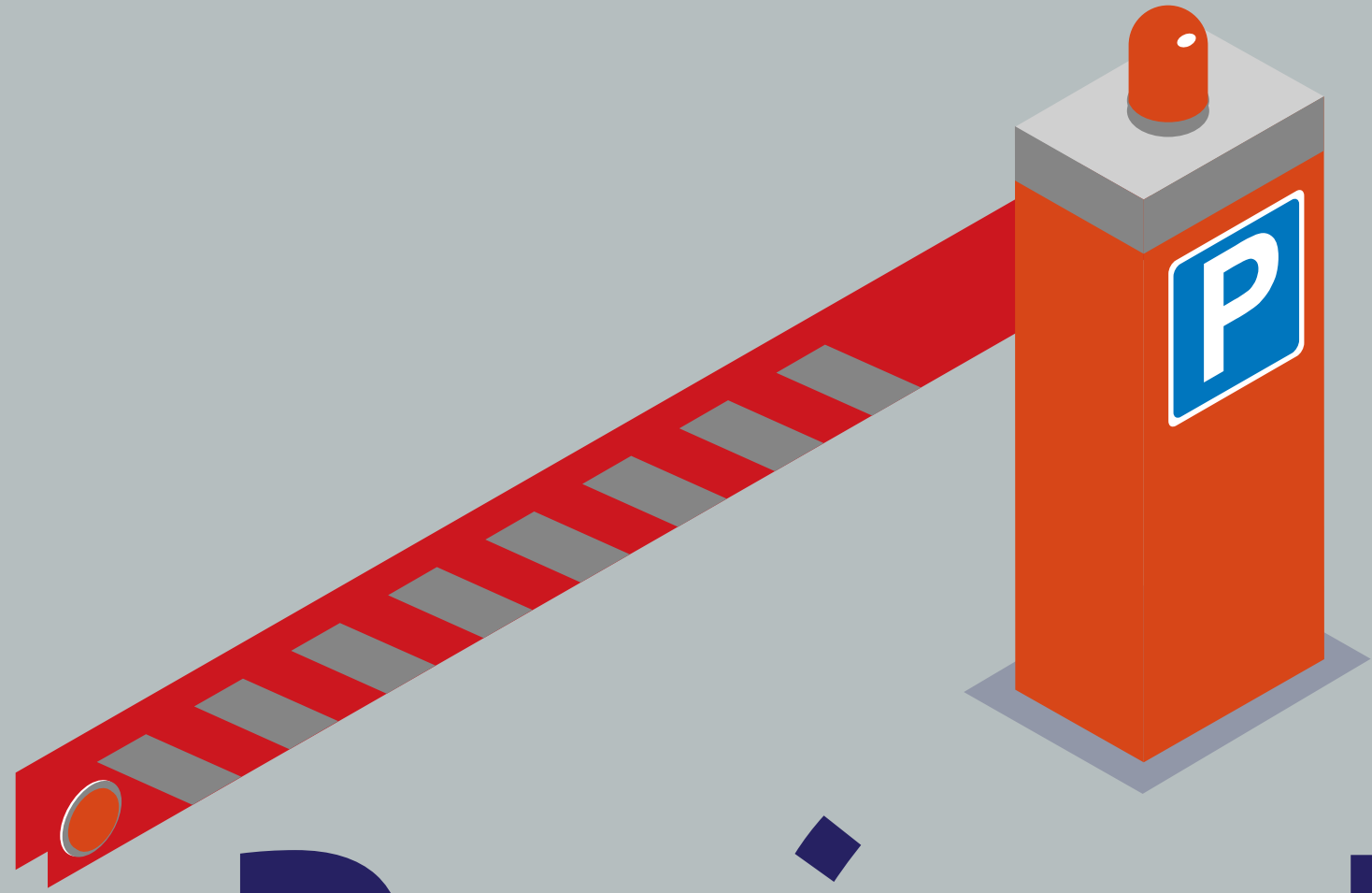
Concurrency, performance, and the backend skills that matter



Growing Market

EV charging and smart parking: the industry of tomorrow





02

Project Context

Core Elements



Existing system & its limitations

- Slow at processing
- Lack of Plate processing
- No Image handling
- Online mode only



10.06.2026 - 22:55 Shift no. 211 - super user

booked articles:

Interim result:

Still due to pay:

Change money:

Please enter article no.:

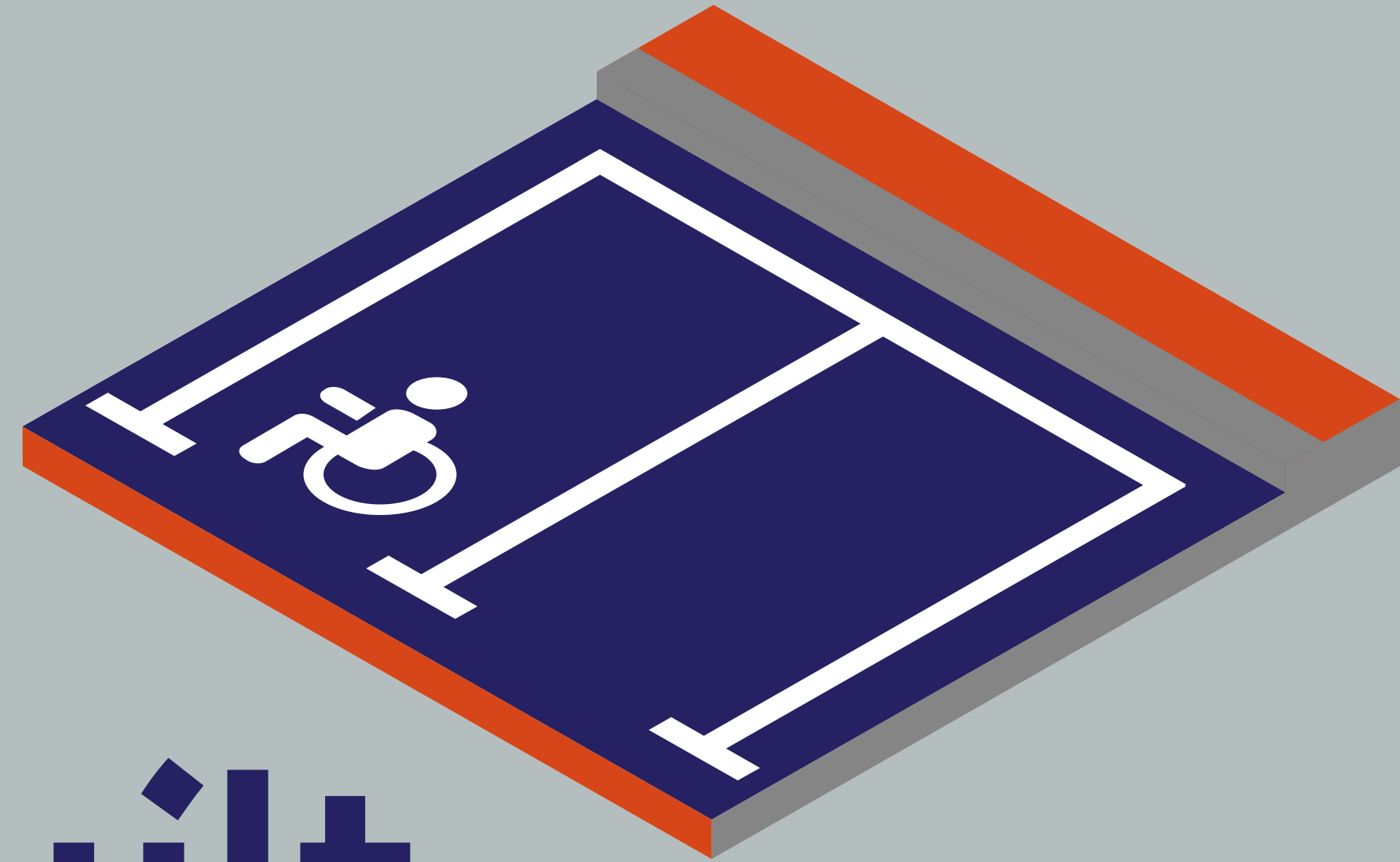
  

702 - ON-LINE -

03

What I Built



POSIFY: The Full Picture



POS System

Cashier-facing interface for payment processing



Backoffice System

Admin dashboard for configuration & monitoring



Printer Service

Handles receipt printing across terminals



Sync Services

Handles data synchronizations between POS terminals and PMS



Integration with barrier Service

Opens and closes gates on payment confirmation



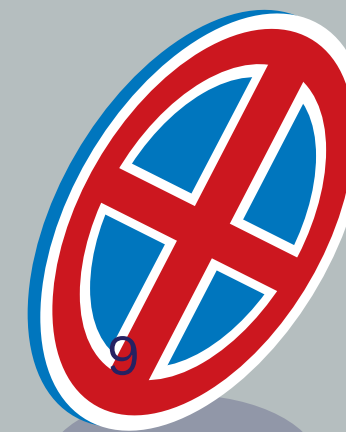
Integration with camera Service

Triggers payment flow by sending detected plate to POS



Integration with tariff Service

Calculates parking fees based on duration and rules

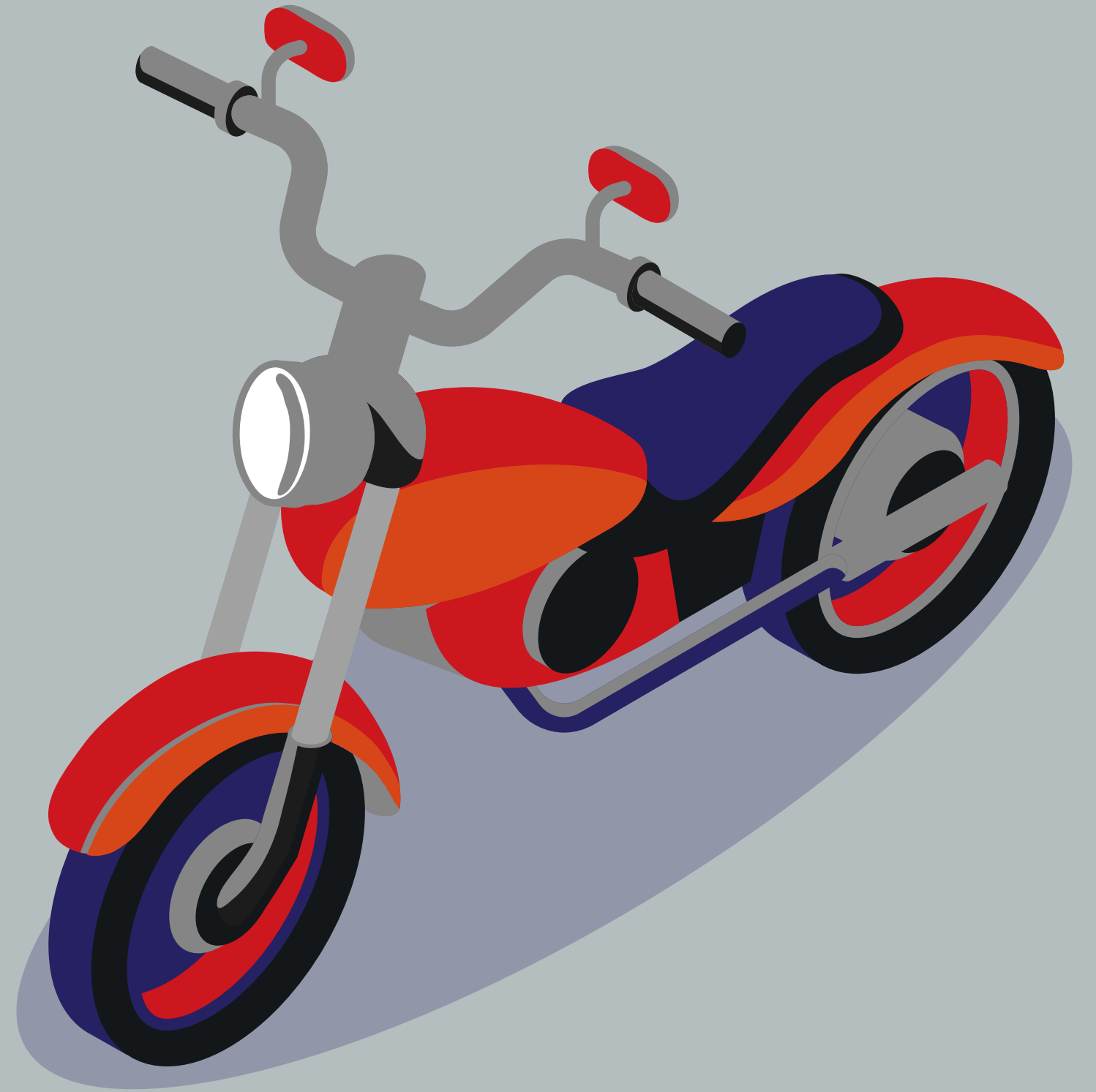


What a client gets?

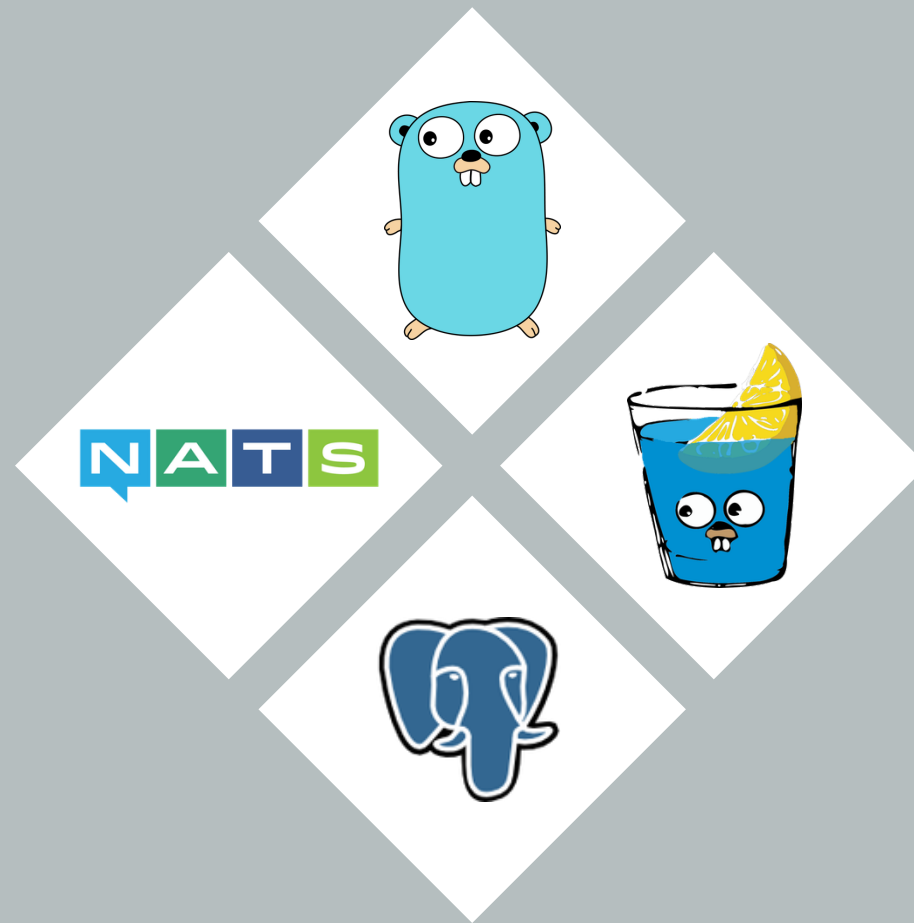
The Problem/The Need	What POSIFY delivers
Slow, outdated cashier interface	Modern POS with real-time payment processing
No license plate recognition	Automatic payment trigger via camera LPN detection
System crashes if network drops	Full offline mode -> operations never stop
No vehicle image at the booth	Cashier sees the vehicle photo on every transaction
Clients need it to fit their existing setup	PMS-agnostic -> plugs into any parking management system
Every parking site has different rules	Fully configurable per site: cameras, articles, terminals

04

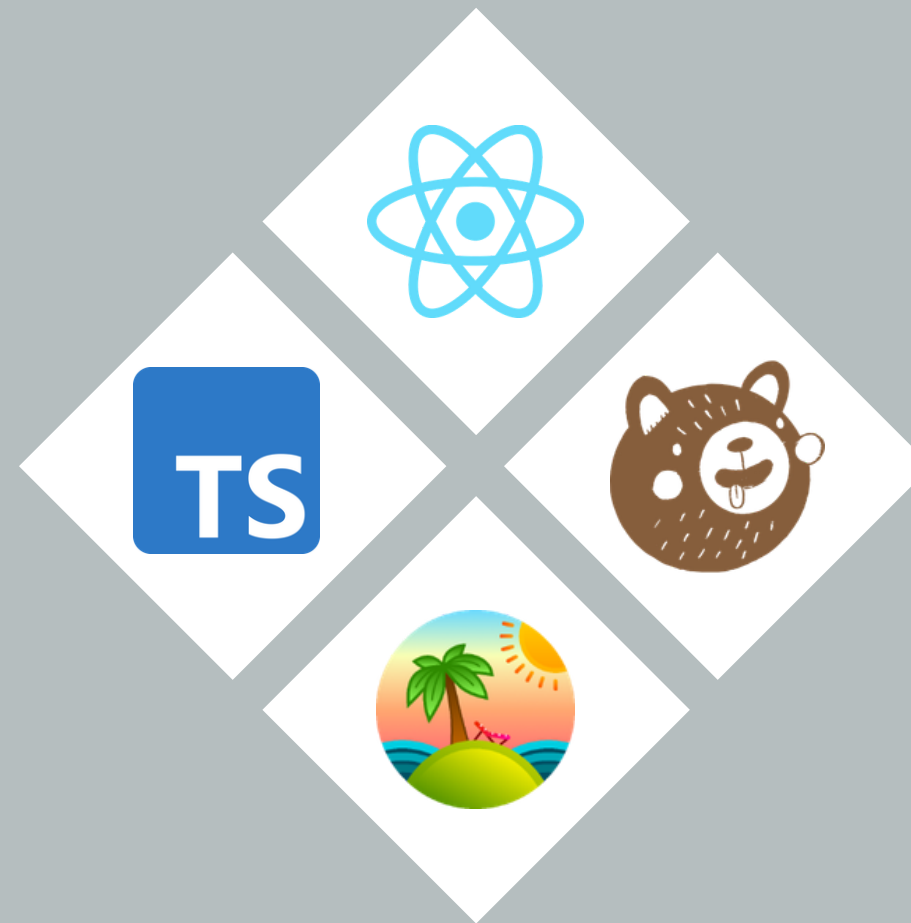
Technical Overview



Technical Overview



Backend Technologies

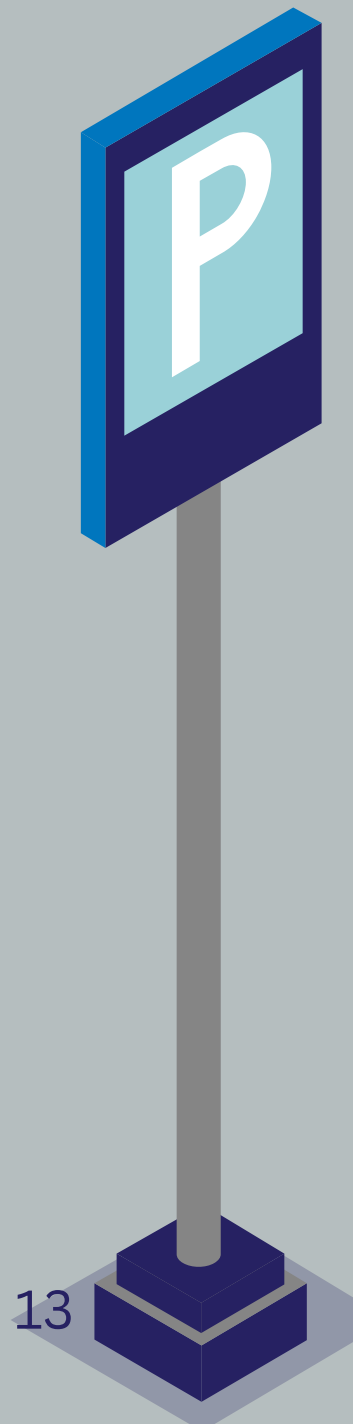
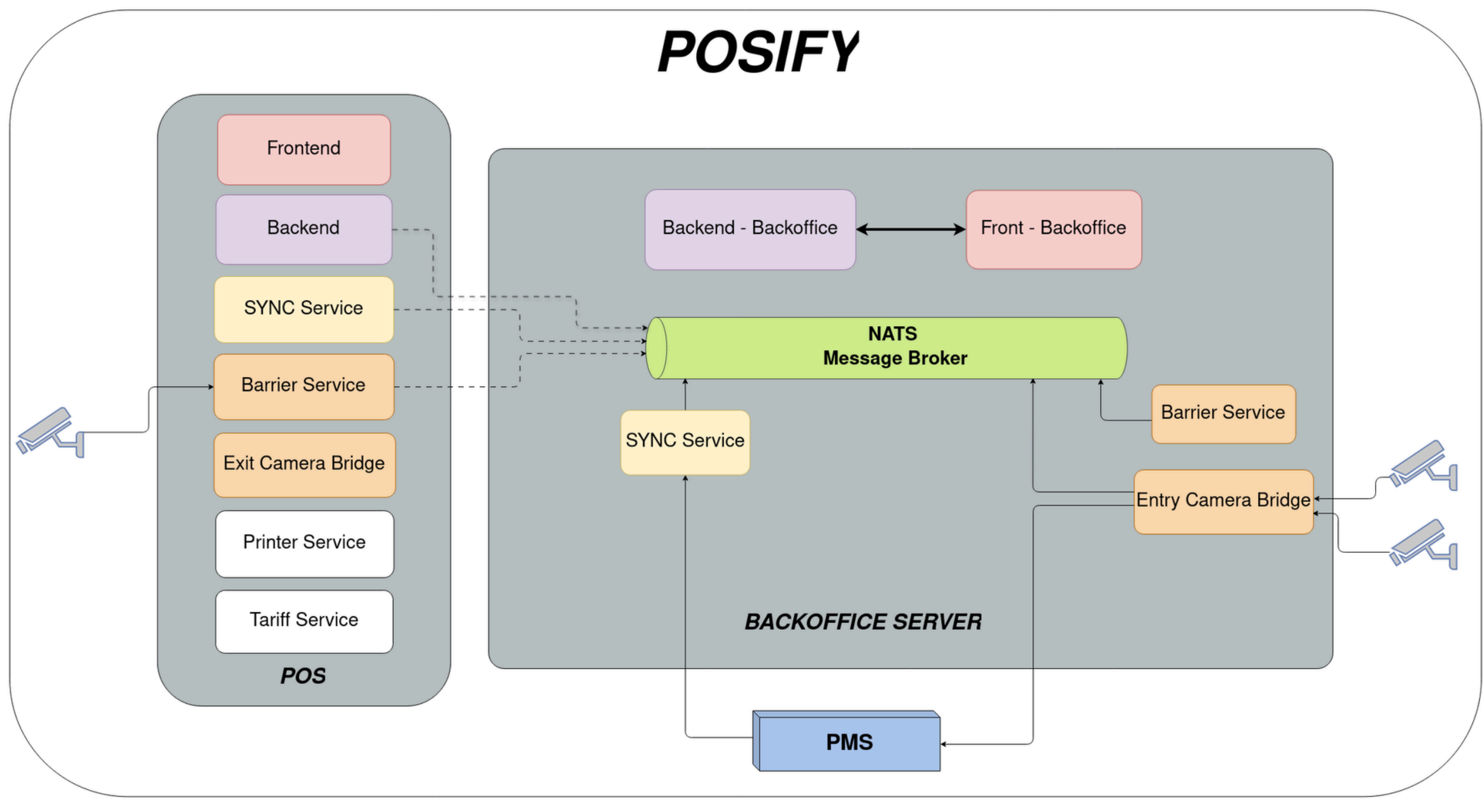


Frontend Technologies



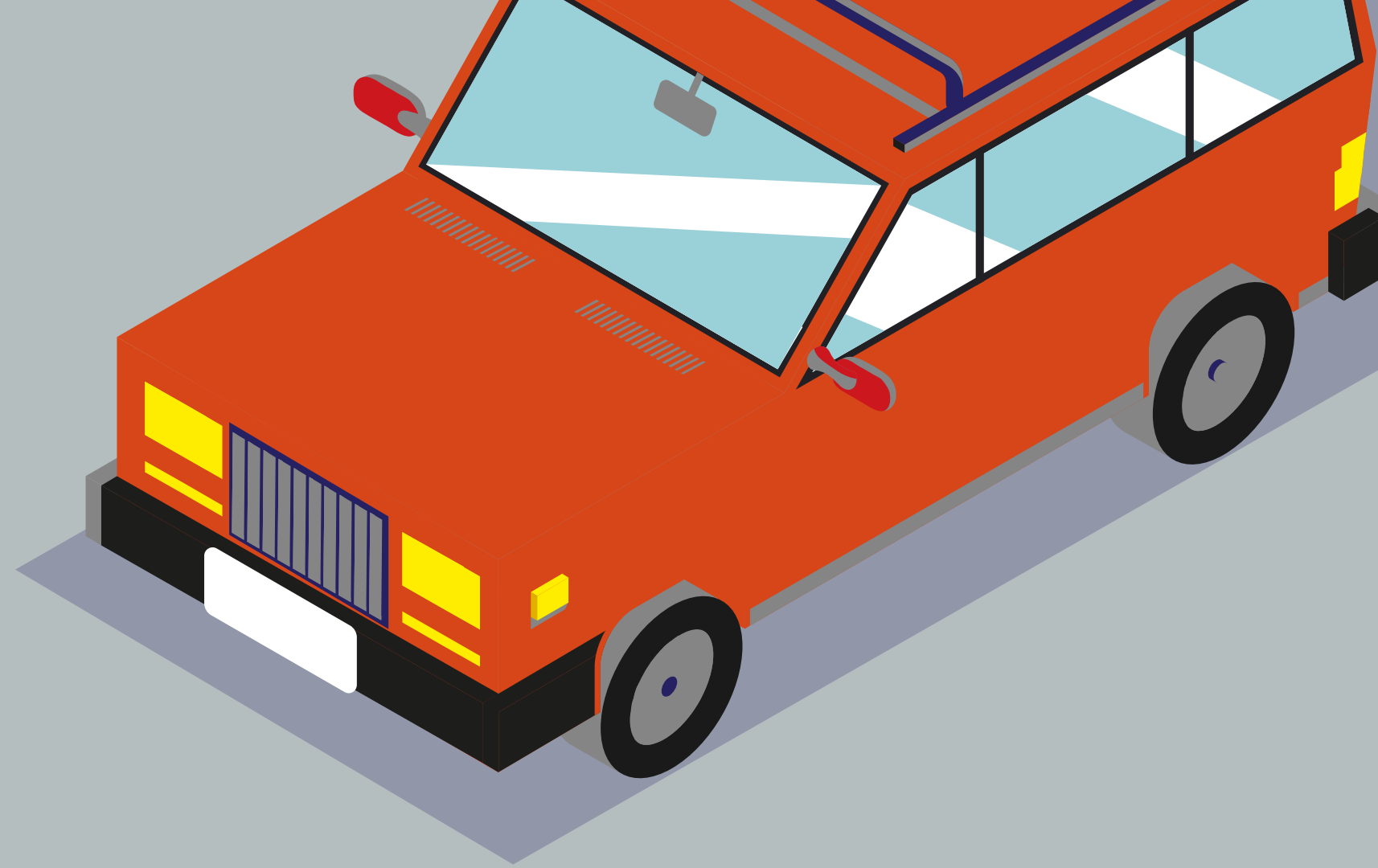
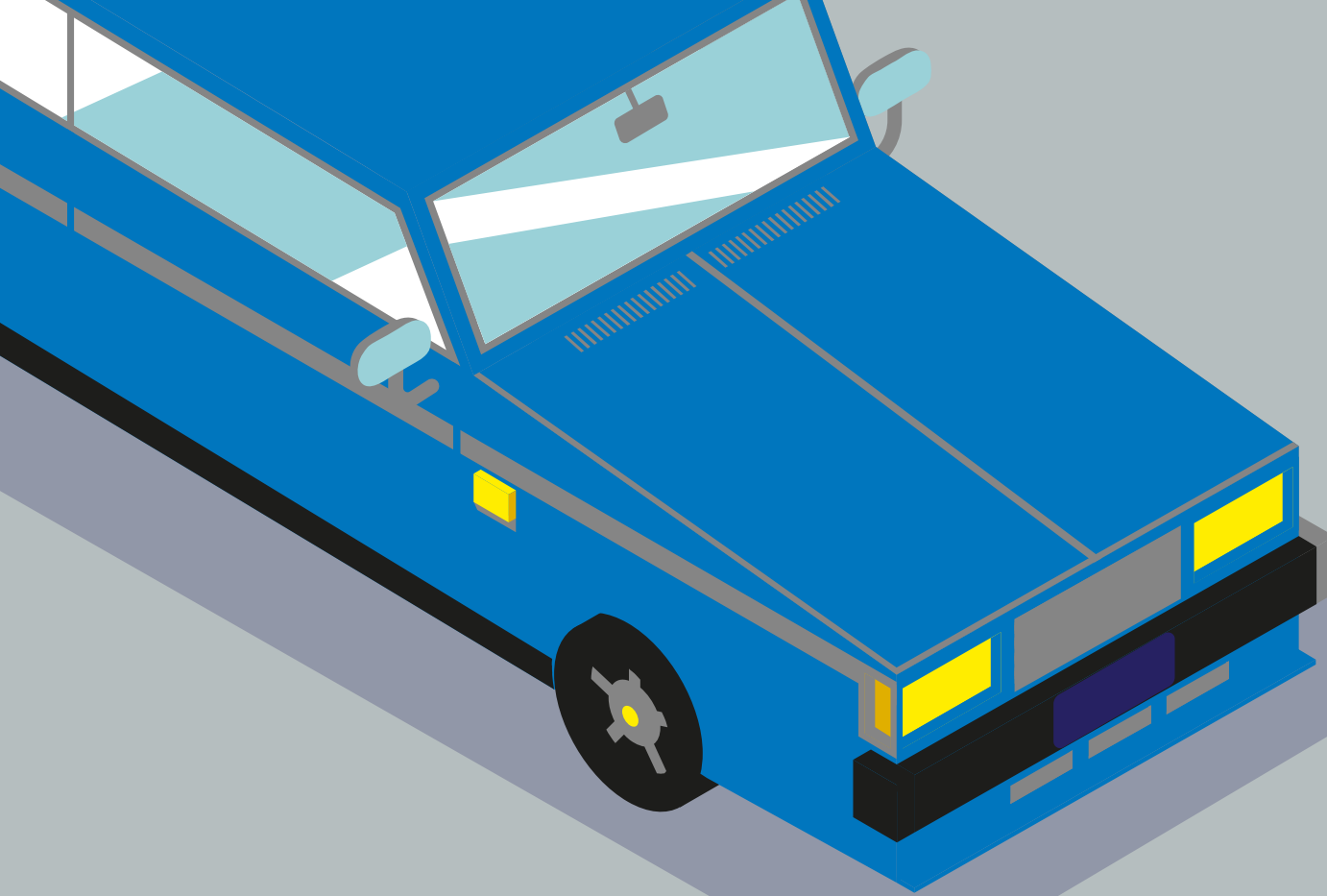
Development Tools &
Infrastructure

System Architecture



05 Demo





06 Conclusion & Perspectives

Conclusion



- ✓ Built a fully functional POS system
- ✓ Delivered a complete Backoffice platform
- ✓ Designed a microservices architecture
- ✓ Integrated with external services
- ✓ Built a PMS-agnostic solution
- ✓ tested, deployed and ready



Perspectives



1. Automated Testing

2. LPN Correction Workflow

3. Additional Operational Scenarios





THANKS



DO YOU HAVE ANY QUESTIONS?

Ayoub Krimi



Internship Project at **Asteroidea** – 2026